

ABSTRACT

Agentic Deep Researcher is an intelligent multi-agent research system designed to automate the process of collecting, analyzing, and presenting information from multiple sources. Traditional search engines mainly provide a list of relevant links, requiring users to manually open, compare, verify, and summarize the information. This project aims to overcome these limitations by developing an AI-powered research assistant that can independently perform research through a structured, multi-step process. The proposed system accepts a broad research question from the user and first uses a Planner Agent to understand the topic and divide it into smaller, meaningful sub-questions. The system then uses Searcher and Fetcher Agents to search the web, retrieve relevant documents or web pages, and collect useful information from multiple sources. The gathered information is analyzed by an AI model and used to generate an initial research draft. A Critic Agent then evaluates the draft for missing information, insufficient source coverage, unsupported claims, and other gaps. If additional information is required, the system automatically initiates another search and retrieval cycle before producing the final response. The final output is a structured research report containing relevant findings, inline citations, source references, and a confidence indication for important claims. The system follows a Plan and Execute approach, allowing the research process to be performed iteratively rather than relying on a single search operation. The project can be implemented using Python, LangGraph, CrewAI, FastMCP, web search APIs, and Large Language Models (LLMs). Live web sources can be used for real-time research, while an optional fixed document corpus can be used for controlled evaluation. The proposed Agentic Deep Researcher can reduce the time and effort required for extensive research while improving the organization, source coverage, and reliability of generated information. The project demonstrates how autonomous AI agents and tool-based workflows can be combined to create a practical research assistant capable of performing complex information-gathering tasks with minimal user intervention.

INTRODUCTION

In recent years, the rapid growth of Artificial Intelligence (AI), Machine Learning, and Large Language Models (LLMs) has significantly changed the way information is searched, processed, and presented. Large amounts of information on the internet has made it easier to find knowledge, but it has also created a new challenge: obtaining accurate, relevant, and well-organized information from multiple sources. Traditional search engines mainly provide a list of webpages or documents related to a query, they are useful for locating information but the user is generally responsible for opening the sources, reading the content, comparing different sources, identifying important information, and preparing the final conclusions. Similarly, conventional AI-based question answering systems can generate useful responses but may not always perform systematic multi-step research across diverse sources. These limitations create a need for an intelligent system performs research in a more structured and iterative manner.

The Agentic Deep Researcher is proposed as an AI-based multi-agent research system that aims to automate the major stages of the research process. Instead of relying on a single AI model to answer a question directly, the system divides the research task among specialized AI agents. The Planner Agent first analyzes the user's research question and divides it into smaller sub-questions. The Searcher Agent then searches for relevant information using appropriate web and academic sources. Tools such as Tavily, Exa, and Semantic Scholar can be used to retrieve information from different types of sources. The Fetcher Agent obtains the relevant content from the selected sources for further processing. After information has been collected, the Critic Agent evaluates the research to determine whether sufficient and relevant evidence has been gathered. If important information is missing or gaps are identified, the system can initiate another research cycle. This iterative process allows the system to refine its research instead of stopping after a single search operation. Technologies such as LangGraph, CrewAI can be used to coordinate the agents and control the research workflow, while Model Context Protocol (MCP) can provide standardized access to external tools. The system can also make use of Retrieval-Augmented Generation (RAG) and knowledge storage techniques to improve the use of retrieved information. Relevant research content can be stored and retrieved when required, helping the language model generate responses based on collected evidence. Finally, the Report Generation module organizes the findings into a structured research report containing relevant explanations, inline citations, source references, and confidence information.

LITERATURE REVIEW

The ReAct: Synergizing Reasoning and Acting in Language Models paper introduced a framework that combines reasoning and action in an iterative process. Instead of generating an answer directly, the model reasons about the task, performs an action such as searching for information, observes the result, and continues reasoning based on the obtained information. This concept is relevant to the proposed system because the Agentic Deep Researcher also follows an iterative research process in which agents reason about the research question and perform search and retrieval actions. However, ReAct mainly focuses on the reasoning-action process rather than a complete multi-agent research workflow.

The Toolformer: Language Models Can Teach Themselves to Use Tools paper demonstrated how language models can make use of external tools and APIs to improve their capabilities. This is important for research applications because an LLM cannot always depend only on its internal knowledge. External tools can provide current and specialized information. This idea supports the integration of tools such as Tavily, Exa, and Semantic Scholar in the proposed system for web and academic information retrieval. However, Toolformer mainly focuses on tool usage and does not provide a complete mechanism for research planning, evaluation, and report generation.

The Reflexion: Language Agents with Verbal Reinforcement Learning paper proposed the use of self-reflection and feedback to improve the performance of language agents. An agent can evaluate its previous result, identify weaknesses, and use this feedback to improve the next attempt. This concept directly supports the Critic Agent in the proposed Agentic Deep Researcher. The Critic Agent evaluates the collected research, identifies missing information or research gaps, and can trigger another search cycle when required. Thus, Reflexion provides the foundation for the iterative refinement aspect of the proposed system.

The AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation paper focused on the development of applications using multiple interacting AI agents. It demonstrates how complex tasks can be divided among specialized agents that communicate and collaborate to achieve a common objective. This concept is reflected in the proposed architecture through specialized Planner, Searcher, Fetcher, and Critic Agents. Each agent performs a specific role while the overall workflow coordinates their activities.

PROBLEM STATEMENT

In today's digital environment, a large amount of information is available through websites, research papers, academic databases, news articles, and other online sources. Although this makes information easily accessible, students, researchers, and professionals often have to search multiple sources, read large amounts of content, compare information, verify its reliability, and manually prepare a final report. This process requires considerable time and effort and may result in incomplete or inconsistent research.

Traditional search engines mainly provide a list of webpages or documents based on keywords. They do not automatically understand the complete research objective, divide a complex question into smaller research tasks, or determine whether sufficient information has been collected. Users must manually visit different sources, identify useful information, and combine the findings. Similarly, conventional AI-based question-answering systems can generate quick responses but may not systematically perform multi-step research, retrieve information from diverse sources, verify the collected evidence, or identify gaps in the research before producing an answer. Managing and comparing this information manually can be difficult. In addition, a single search operation may not provide enough information to answer a broad research question. Therefore, a research system should be capable of performing additional searches when the collected information is insufficient.

To address these problems, there is a need for an intelligent and automated research system that can understand a user's research question, divide it into meaningful sub-questions, search relevant sources, retrieve useful information, evaluate the collected results, and refine the research when necessary. The system should also organize the collected information into a structured report and provide citations so that users can understand the sources supporting the generated information.

The proposed Agentic Deep Researcher aims to address these challenges through a multi-agent architecture. A Planner Agent analyzes and decomposes the research question, while Searcher and Fetcher Agents locate and retrieve relevant information from external sources. A Critic Agent evaluates the collected information and identifies research gaps. If the information is insufficient, the system can perform additional research before generating the final output. The system is therefore designed to reduce manual research effort and provide a more systematic, iterative, and transparent approach to deep research.